SEA-HELM Audio: Evaluating Multilingual ASR-LLM Pipeline Performance Across SEA-HELM Subsets

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Abstract

We introduce **SEA-HELM Audio**, a Southeast Asian (SEA) multilingual evaluation framework that extends the text-based SEA-HELM benchmark to assess performance degradation in cascaded automatic speech recognition and large language model (ASR–LLM) pipelines. Using synthesized speech, we compare speech-input and text-input performance across five languages—Chinese, English, Malay, Indonesian, and Tamil—on multiple-choice reasoning and instruction-following tasks. We evaluate open-weight and commercial ASR–LLM systems using identical verifiers and quantify the **speech gap** as the difference between text and speech accuracy. Results show substantial degradation, with accuracy drops of 9.00–18.50 percentage points for multiple-choice reasoning and 20.00–42.86 points for instruction following. Tamil exhibits the largest degradation, exceeding 40 percentage points. Our findings highlight the limitations of text-only benchmarks in predicting voice-agent reliability and demonstrate the value of SEA-HELM Audio for evaluating multilingual voice-agent systems.

Index Terms: automatic speech recognition, multilingual evaluation, spoken language understanding, SEA-HELM Audio

1. Introduction

As artificial intelligence systems transition from text-based interfaces to conversational voice-enabled agents, evaluating the end-to-end accuracy of cascaded speech-language stacks has become a critical research priority. While text-based Large Language Models (LLMs) continue to achieve high benchmark scores, deploying these models within voice pipelines introduces acoustic transcription errors, phonetic ambiguities, and domain-specific degradation. Quantifying this degradation, specifically measuring the accuracy penalty a user incurs by speaking rather than typing, is essential for understanding real-world agent reliability across diverse tasks.

Existing spoken language model benchmarks, such as VoiceBench [1], evaluate capabilities including speech recognition, reasoning, and instruction following through spoken inputs. However, their predominantly English-language evaluation settings provide limited insight into how voice-agent systems perform on Southeast Asian languages and tasks. In particular, it remains unclear to what extent transcription errors alter task-critical information and how these errors affect downstream task performance, especially in the context of Southeast Asian languages.

To address the need for localized evaluation, AI Singapore developed **SEA-HELM** (Southeast Asian Holistic Evaluation of Language Models) [2]. SEA-HELM provides authentic per-language prompts and tailored programmatic verifiers across regional languages. However, SEA-HELM was engineered natively as a text-only benchmark, consisting exclusively of written prompts without paired audio recordings.

This paper addresses the central research question: **How can a text-based, regionally grounded benchmark be adapted into a speech-based evaluation dataset to systematically measure the impact of speech input on downstream LLM task performance across Southeast Asian languages?** By bridging SEA-HELM's text prompts with synthesized audio pipelines, we establish a framework to quantify the **Speech gap**, the difference between $\mathrm{Score}_{\mathrm{text}}$ and $\mathrm{Score}_{\mathrm{speech}}$, across multiple Southeast Asian languages, providing empirical baselines for regional speech agent deployment.

2. Methodology

This section outlines the experimental framework, dataset selection, and evaluation metrics employed to systematically isolate and quantify the performance degradation introduced by the speech modality across our five target languages.

2.1. Experimental Design

Prior to finalizing the evaluation pipeline, we conducted an exploratory phase to establish a robust and regionally relevant testing environment. A primary distinction between our setup and existing voice evaluation frameworks lies in audio corpus generation: VoiceBench uses pre-recorded human speech, whereas SEA-HELM provides only text prompts. Consequently, our cascade architecture synthesizes its own input speech before evaluating downstream models.

To ensure audio fidelity across regional languages, we generated synthetic speech with OmniVoice [3] Text-to-Speech (TTS) engines, selected for acoustic naturalness, prosodic accuracy, and regional accent representation.

For task design, we selected multiple-choice and instruction-following formats, which are established in voice evaluation, and instantiated them with SEA-HELM text prompts. This approach preserves regional nuances, code-switching patterns, and local cultural contexts across Chinese, Malay, Tamil, Indonesian, and English. By evaluating each sample twice, once through the synthesized voice pipeline ($\mathrm{Score}_{\mathrm{speech}}$) and once via direct text prompts ($\mathrm{Score}_{\mathrm{text}}$), we establish a controlled baseline to isolate and measure the precise performance penalty imposed by the speech-to-text layer.

2.2. Datasets and Task Selection

To evaluate the voice cascade cost-effectively and deterministically, we selected tasks from SEA-HELM that rely on **rule-based scoring** and closely align with established VoiceBench

task types.

The evaluation suite draws from four primary SEA-HELM task subsets, summarized in Table 1.

2.2.1. Benchmark Alignment and Active Subset Scope

For the initial empirical runs detailed in this study, benchmarking focuses on the SEA-HELM IFEval and Global-MMLU-Lite subsets across our five focus languages (Chinese, Malay, Tamil, Indonesian, and English).

The selection criteria prioritized two key attributes:

1. **Speakable Prompts:** Prompts can be read aloud as natural speech without removing formatting or template markup first.
2. **Verifiable Evaluation:** All scoring is executed via programmatic, rule-based verifiers, so the text path and the speech path are scored in exactly the same way.

2.3. Evaluation Metrics

To formalize the measurement of the speech modality penalty, the following metrics were captured across every benchmark run:

- **Text accuracy** ($\mathrm{Score}_{\mathrm{text}}$)**:** The accuracy of the target LLM when provided with the original, ground-truth text input. This represents the theoretical performance ceiling for the task.
- **Speech accuracy** ($\mathrm{Score}_{\mathrm{speech}}$)**:** The end-to-end accuracy of the target LLM when the prompt is spoken, synthesized by TTS and transcribed by ASR.
- Δ **(The Gap):** $\Delta = \mathrm{Score}_{\mathrm{text}} - \mathrm{Score}_{\mathrm{speech}}$. This absolute difference represents the modality penalty, quantifying how much task performance degrades due to transcription errors, phonetic ambiguities, or missing acoustic signals.

3. SEA-HELM Audio

SEA-HELM contains only written prompts, so SEA-HELM Audio creates the spoken input itself: each prompt is read aloud by a speech synthesizer, transcribed by the ASR system under test, and answered by the LLM. Comparing this against the same LLM answering the original text isolates the cost of the speech front-end.

3.1. Evaluation Flow

Each SEA-HELM prompt is evaluated in two ways (Figure 1). In the **speech path**, the prompt is converted to speech with TTS, transcribed back to text by ASR, and the transcript is answered by the LLM. In the **text path**, the LLM answers the original prompt directly. Both answers are scored by the same rule-based verifier, giving $\mathrm{Score}_{\mathrm{speech}}$ and $\mathrm{Score}_{\mathrm{text}}$. Because the two paths share the same LLM and the same verifier, the difference between them ($\Delta = \mathrm{Score}_{\mathrm{text}} - \mathrm{Score}_{\mathrm{speech}}$) is the accuracy lost by speaking the prompt instead of typing it.

3.2. Spoken Prompt Construction

The TTS system plays the role of the user. Its voice is chosen independently of the system being evaluated, so every ASR–LLM configuration hears the same spoken prompts and differences in $\mathrm{Score}_{\mathrm{speech}}$ reflect the system rather than the input.

Only the content a user would actually say is synthesized. For multiple-choice tasks, this is a short instruction followed by the question and its four options; for instruction-following tasks, it is the prompt exactly as written. Written conventions that do not read naturally aloud are adapted per language: in Chinese, the option labels A–D and the answer marker are written as Chinese words. The answer key (the correct option, or the constraints an IFEval response must satisfy) is never spoken and is used only at scoring time.

3.3. Speech Recognition

The ASR system turns the synthesized speech back into text. We evaluate three systems chosen for their coverage of the target languages (Table 2): Qwen3-ASR-1.7B [7] as an open-weight multilingual baseline, MERaLiON3-ASR-3B [8] as a regionally adapted open-weight model for Tamil, and ElevenLabs Scribe Realtime as a commercial service. Open-weight models use greedy decoding so that transcripts are reproducible.

Transcription quality is also measured directly, as Word Error Rate (WER) and Character Error Rate (CER) between the transcript and the original prompt. This separates how much the ASR changed the prompt from how much those changes affected the task.

3.4. Response Generation

The LLM answers the ASR transcript (speech path) and the original prompt (text path). We evaluate SEA-LION-4.5-27B [9], a regional open-weight model, and Gemini 3.5 Flash, a commercial model. Both use deterministic decoding (temperature 0), so any difference between the two paths comes from the input rather than from sampling.

3.5. Scoring

All responses are scored with rule-based verifiers taken from SEA-HELM, which keeps scoring deterministic, cheap to repeat, and identical across the two paths. For multiple-choice questions, the chosen option letter (A–D) is extracted from the response and compared with the answer key. For instruction following, each response is checked against the constraints stated in its prompt and is counted as correct only if it also uses the expected language.

4. Results

This section presents the empirical findings from our multilingual voice cascade evaluations. Performance is reported across two primary axes: task type (multiple-choice questions via Global-MMLU-Lite vs. instruction following via SEA-HELM IFEval) and pipeline configuration (Regional Open-Weight Baseline vs. Commercial Stack).

4.1. Multiple Choice Reasoning (Global-MMLU-Lite)

Evaluations on the SEA-HELM Global-MMLU-Lite subset were conducted for Chinese and English, assessing general domain knowledge under a multiple-choice format (400 samples per run). Across both languages, the commercial stack (ElevenLabs Scribe Realtime paired with Gemini 3.5 Flash) outperformed the regional open-weight baseline in both text accuracy and speech accuracy.

However, Chinese demonstrated a distinct resilience to the modality shift compared to English. For Chinese, Δ remained between 9.00 and 10.75 points, whereas English incurred a much steeper penalty of 15.50 to 18.50 points.